

IN-HOUSE PROJECT REPORT
ON
Disease Prediction using Patient Treatment History and Health Data

IV Semester

BACHELOR OF COMPUTER APPLICATIONS
BCA (A)

By

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Under the esteemed guidance of

P. VIKAS



DEPARTMENT OF COMPUTER SCIENCE

P.B. SIDDHARTHA COLLEGE OF ARTS AND SCIENCE

VIJAYAWADA-10

(An Autonomous college under the jurisdiction of Krishna University Machilipatnam
A college with potential for excellence, re-accredited at the level 'A+' by NAAC, Bangalore)

2025-2026

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CERTIFICATE

This is to clarify that IN-HOUSE PROJECT entitled " **Disease Prediction using Patient Treatment History and Health Data**", Is Bonafide work of **Abdul Sahel (244153P)** submitted in partial fulfilment of the requirement for the award of degree of **BACHELOR OF COMPUTER APPLICATIONS** from P.B. Siddhartha college of Arts and Science, Vijayawada.

Project Guide

Head of the Department

Examiner

DECLARATION

I hereby declare that the In-House Project (IHP) report entitled " **Disease Prediction using Patient Treatment History and Health Data** " submitted by me in partial fulfilment of the requirement for the award of degree in BACHELOR OF COMPUTER APPLICATIONS in the record of work originally carried out by from 27-April 2026 – 27 May 2026 under the guidance of **P. VIKAS** FIXITY EDX, Vijayawada.

Abdul Sahel
(244153P)

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Abdul Sahel
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ABSTRACT

The rapid growth of healthcare data and advancements in machine learning have created opportunities for developing intelligent systems capable of assisting in medical diagnosis and disease prediction. This project, titled “Disease Prediction using Patient Treatment History and Health Data,” focuses on building a machine learning-based disease prediction system using symptom-related healthcare data. The primary objective of the project is to predict diseases accurately based on patient symptoms and treatment history using supervised machine learning classification techniques.

The dataset used in this project contains multiple symptom-based features and disease labels. The preprocessing stage includes data cleaning, removal of unnecessary columns, feature selection, and train-test splitting for model development. Exploratory Data Analysis (EDA) techniques and graphical visualizations are used to understand the distribution of symptoms and disease categories within the dataset. Various machine learning algorithms such as Random Forest Classifier, Decision Tree Classifier, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) are implemented and compared to identify the best-performing model.

The project uses the Scikit-learn library for model building and evaluation, while Pandas, NumPy, Matplotlib, and Seaborn are used for data manipulation and visualization. Model performance is evaluated mainly using classification accuracy metrics. Among the tested algorithms, the Random Forest Classifier achieved the best performance with higher prediction accuracy and better generalization capability compared to other classifiers.

The developed system demonstrates that machine learning can effectively analyze healthcare symptom data and provide reliable disease predictions. This approach can help healthcare professionals make faster decisions and support early disease identification. The project also highlights the importance of preprocessing, feature handling, and model comparison in achieving better prediction performance.

Overall, this project presents a complete machine learning pipeline including data preprocessing, exploratory analysis, model training, evaluation, and result visualization. The proposed disease prediction system can serve as a foundation for future intelligent healthcare applications and can be further enhanced using larger datasets, advanced algorithms, and real-time deployment technologies.

CONTENTS

S No.	Topics	Page No.
1.	Introduction	1-2
2.	System Requirements	3-4
3.	Software Environment	5-6
4.	Project pre-requisites	7-9
5.	Source Code	10-19
6.	Output Screen / Results	20-24
7.	Conclusion	25-26
8.	Bibliography / References	27-28

Introduction

The healthcare industry is increasingly adopting Machine Learning (ML) techniques to improve disease diagnosis and patient care. Early and accurate disease prediction plays a significant role in reducing medical costs, assisting doctors in clinical decision-making, and providing timely treatment to patients. Traditional diagnosis methods often depend on manual observation and medical expertise, which can be time-consuming and prone to errors when dealing with large amounts of patient information. Therefore, developing an intelligent disease prediction system using patient treatment history and health-related data has become an important research area in modern healthcare analytics.

This project, titled “**Disease Prediction using Patient Treatment History and Health Data**,” focuses on building a machine learning model capable of predicting diseases based on symptoms and patient-related health information. The notebook aims to solve the problem of identifying the most probable disease from a set of symptoms provided in the dataset. The project uses a supervised machine learning approach because the target variable, known as “**prognosis**,” is already labeled in the dataset.

The dataset used in this project consists of approximately **4,920 rows and 134 columns** in the training data, where most columns represent symptoms or health conditions, and one column represents the target disease category. The testing dataset is used to evaluate the performance of the trained models. Some important features used in prediction include symptoms such as **fever, headache, fatigue, vomiting, and cough**. The target variable is the **disease/prognosis** that the system attempts to predict.

To achieve accurate disease prediction, multiple classification algorithms are implemented and compared in the notebook. These algorithms include **Random Forest Classifier, Decision Tree Classifier, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM)**. Since the project deals with predicting predefined disease categories, it is classified as a **Classification Problem** under supervised learning.

The main objective of this project is to compare the performance of different machine learning models and identify the algorithm with the highest prediction accuracy. The success metric used in the notebook is **classification accuracy**, calculated using the `accuracy_score` metric from Scikit-learn. The remaining sections of the documentation discuss dataset preprocessing, model implementation, performance evaluation, visualization of results, and final conclusions derived from the experiment.

Additionally, the project emphasizes the importance of healthcare data analysis and demonstrates how machine learning can support medical professionals in making faster and more accurate decisions. By analyzing patient symptoms systematically, the developed model reduces the complexity involved in manual disease identification and improves prediction efficiency. The implementation of multiple classification algorithms also helps in understanding how different machine learning techniques perform on healthcare

datasets. Comparing these algorithms provides valuable insights into model accuracy, training behavior, and prediction capability.

The project workflow includes several important stages such as data collection, preprocessing, exploratory data analysis, model training, testing, and performance comparison. During preprocessing, unnecessary columns are removed and the dataset is prepared for machine learning operations. Exploratory Data Analysis (EDA) is performed using graphical visualizations to better understand symptom distribution and disease frequency. These visualizations improve interpretability and help identify patterns within the dataset.

Another important contribution of this project is the use of visualization techniques for comparing algorithm performance. Accuracy comparison graphs and distribution charts provide a clear understanding of model behavior and help identify the most suitable classifier for disease prediction. The project also demonstrates the practical application of Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn in healthcare analytics.

The developed system can be further extended into a real-time healthcare application capable of assisting hospitals, clinics, and diagnostic centers. With larger datasets and advanced machine learning techniques, the prediction accuracy can be improved further. Therefore, this project serves as a strong foundation for future intelligent healthcare systems and medical decision-support applications.

System Requirements

The proposed **Disease Prediction using Patient Treatment History and Health Data** project is implemented using machine learning algorithms in a Jupyter Notebook/Google Colab environment. Since the dataset used in the project is relatively small and structured, the computational requirements are moderate and suitable for standard personal computers or cloud-based notebook platforms. The project mainly involves data preprocessing, model training, prediction, and evaluation using classification algorithms such as Random Forest, Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM).

Hardware Requirements: -

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3 / AMD equivalent	Intel Core i5/i7 or higher
RAM	4 GB	8 GB or higher
Storage	500 MB free space	1 GB or higher
GPU	Optional	Not required for this project
Display	1366 × 768 resolution	Full HD (1920 × 1080)

The dataset contains approximately **4,920 samples and 134 features**, occupying only a few megabytes of storage. Since the notebook uses traditional machine learning models from Scikit-learn and does not involve deep learning frameworks such as TensorFlow-GPU or PyTorch CUDA, GPU acceleration is not necessary.

Software Requirements: -

Software	Requirement
Operating System	Windows, Linux, or macOS
Programming Language	Python 3.8 or above
Development Environment	Jupyter Notebook or Google Colab
Libraries Required	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
GPU Support	Not applicable

Runtime Considerations: -

- **Data Loading:** Approximately 5–10 seconds
- **Data Preprocessing:** Approximately 10–20 seconds
- **Model Training:** Approximately 20–40 seconds
- **Model Evaluation and Visualization:** Approximately 10–15 seconds
- **Total Execution Time:** Approximately 1–2 minutes

Recommended Setup: -

For this project, **Google Colab** is the recommended platform because it provides a pre-configured Python environment with all major machine learning libraries already installed. It also eliminates dependency and installation issues faced in local systems. However, the project can also be executed efficiently on a local machine using Jupyter Notebook with moderate hardware specifications.

Software Environment

The **Disease Prediction using Patient Treatment History and Health Data** project is developed using Python-based machine learning libraries in a Jupyter Notebook/Google Colab environment. The software environment provides tools for data preprocessing, visualization, model training, and performance evaluation. Since the project uses traditional supervised machine learning techniques, lightweight libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn are sufficient for efficient implementation.

Import Statements Used in the Project

```
import NumPy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.metrics import accuracy_score
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.svm import SVC
```

Data Processing Tools

The project mainly uses **NumPy** and **Pandas** for handling and preprocessing the dataset. **NumPy** is a numerical computing library used for performing mathematical and array-based operations efficiently. It helps in handling multidimensional arrays and supports faster numerical computations required during model training.

Pandas is used for reading, cleaning, and manipulating tabular healthcare data. In this project, it is specifically used for loading CSV files, checking missing values, dropping unnecessary columns, and preparing data for machine learning algorithms. Functions such as `read_csv()`, `drop()`, `head()`, `info()`, and `shape` are commonly used. The notebook also uses preprocessing operations like removing the Unnamed: 133 column and splitting features and target variables.

ML and Model Training Tools

The project uses the **Scikit-learn** library, one of the most popular machine learning frameworks in Python. Scikit-learn provides simple and efficient implementations of supervised learning algorithms. Since the project focuses on disease prediction, classification models are used.

The notebook implements multiple algorithms including:

- **Random Forest Classifier** – Used for ensemble-based disease prediction with high accuracy.
- **Decision Tree Classifier** – Helps create a rule-based prediction structure.
- **K-Nearest Neighbors (KNN)** – Predicts diseases based on similarity between patient records.
- **Support Vector Machine (SVM/SVC)** – Used for separating disease categories using hyperplanes.

Functions such as `train_test_split()` are used for dividing data into training and testing datasets, while model methods like `fit()` and `predict()` are used for training and prediction.

Evaluation and Metrics Tools

The project evaluates model performance using the **`accuracy_score()`** function from Scikit-learn. Accuracy is the primary metric because the task is a classification problem. The notebook compares the prediction accuracy of all implemented algorithms to identify the best-performing model for disease prediction.

Visualization Tools

For graphical analysis and result visualization, the notebook uses **Matplotlib** and **Seaborn**. **Matplotlib** is used to generate bar graphs and performance comparison charts for machine learning models. **Seaborn** enhances data visualization with improved statistical plots and styling.

The visualizations mainly include:

- Accuracy comparison bar charts
- Dataset distribution plots
- Feature-related visual analysis

IDE and Development Environment

The project is developed in **Jupyter Notebook/Google Colab**, which provides an interactive coding environment suitable for machine learning experiments. Google Colab is particularly useful because it offers free cloud-based execution, pre-installed ML libraries, and easy dataset management. It also simplifies visualization and experimentation without requiring complex local installations.

Project pre-requisites

Before implementing the **Disease Prediction using Patient Treatment History and Health Data** project, users should possess basic knowledge of machine learning concepts, Python programming, and data analysis techniques. Since this project is based on supervised machine learning classification algorithms, understanding the fundamental concepts of classification problems and healthcare datasets is important for successful implementation and interpretation of results.

Prior Knowledge Required

The project requires a basic understanding of **Python programming fundamentals**, including variables, loops, functions, arrays, and object-oriented concepts. Knowledge of **NumPy** and **Pandas** is essential because these libraries are used for data manipulation, preprocessing, and handling tabular datasets.

Students should also understand:

- Data visualization concepts using graphs and charts
- Statistical basics such as mean, standard deviation, and data distribution
- Classification concepts including class labels, decision boundaries, prediction accuracy, and probability
- Train-test data splitting and model validation techniques

The notebook implements multiple machine learning algorithms, including:

- **Random Forest Classifier**
- **Decision Tree Classifier**
- **K-Nearest Neighbors (KNN)**
- **Support Vector Machine (SVM)**

Therefore, learners should have introductory knowledge of how these algorithms work and how they are used for predictive modeling.

Technical Skills Needed

To successfully execute this project, users should know:

- Exploratory Data Analysis (EDA)
- Data cleaning and preprocessing

- Handling missing or unnecessary columns
- Feature selection and target variable separation
- Training machine learning models
- Evaluating models using accuracy metrics
- Visualizing model performance using graphs

The project mainly works with **numerical and binary symptom-based features** where symptoms are represented as columns and diseases are represented as target labels.

Required Installations

Install the following Python libraries before running the notebook:

```
pip install NumPy
pip install pandas
pip install matplotlib
pip install seaborn
```

pip install scikit-learn **Note:** For Google Colab users, these libraries are already pre-installed and no manual installation is required.

Dataset Setup

- **Dataset Name:** Disease Prediction Dataset
- **File Format:** CSV
- **Dataset Size:** Approximately 4,920 rows $\times$ 134 columns
- **Source:** Healthcare symptom and disease prediction dataset used for academic machine learning analysis
- **Placement:** Store dataset files in the same directory as the notebook

Dataset loading example:

```
import pandas as pd
df = pd.read_csv('Training.csv')
```

Environment Setup Steps

1. Open the notebook in **Google Colab** or **Jupyter Notebook**
2. Install required libraries if needed
3. Upload or place dataset files in the working directory
4. Run the first import cell to verify library installation
5. Execute dataset loading cells and confirm data is displayed correctly

Hardware Recommendations

- **Minimum RAM:** 4 GB
- **Recommended RAM:** 8 GB
- **Processor:** Intel Core i3 or above
- **Storage Requirement:** Minimum 500 MB free space

Since the dataset is relatively small and the project uses lightweight Scikit-learn algorithms, a standard personal computer is sufficient for smooth execution.

Source Code

Project Structure Overview

The notebook for the **Disease Prediction using Patient Treatment History and Health Data** project follows a structured machine learning workflow. The code is organized into multiple stages beginning with library imports and dataset loading, followed by exploratory data analysis (EDA), data preprocessing, model development, evaluation, and visualization of results. The overall workflow can be summarized as:

Data Loading → Data Exploration → Data Preprocessing → Model Training → Model Evaluation → Result Visualization

The notebook uses supervised machine learning classification algorithms to predict diseases based on symptom-related features. Multiple classification models are trained and compared using accuracy metrics to determine the best-performing algorithm.

Data Loading and Exploration

The first stage of the notebook imports required libraries and loads the training and testing datasets into Pandas DataFrames.

```
import NumPy as np
import pandas as pd
```

```
train=pd.read_csv('/content/Training.csv')
test=pd.read_csv('/content/Testing.csv')
```

The `read_csv()` function is used to load the datasets from CSV files. The training dataset contains symptom-related features and disease labels, while the testing dataset is used to evaluate model performance.

The notebook then performs basic dataset inspection using:

```
train.head()
test.head()
```

These commands display the first five rows of the datasets and help verify whether the data has loaded correctly. Additional exploratory methods such as `shape`, `info()`, and checking column names help understand the dataset structure, number of features, and target variable.

Exploratory Data Analysis (EDA)

The notebook performs basic visualization to understand the class distribution of diseases present in the dataset. The primary visualization used is a bar chart representing the frequency of each disease category.

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(18,6))
```

```
train['prognosis'].value_counts().plot(kind='bar')
```

```
plt.title('Class Distribution of Prognosis')
```

```
plt.xlabel('Disease')
```

```
plt.ylabel('Count')
```

```
plt.xticks(rotation=90)
```

```
plt.show()
```

Explanation

```
import matplotlib.pyplot as plt
```

This line imports the **Matplotlib** library, which is used for creating graphs and visualizations in Python.

```
plt.figure(figsize=(18,6))
```

Creates a graph figure with a width of 18 inches and height of 6 inches.

This larger size helps display all disease names clearly.

```
train['prognosis'].value_counts().plot(kind='bar')
```

- `train['prognosis']` selects the target disease column.
- `value_counts()` counts how many times each disease appears.
- `plot(kind='bar')` creates a bar graph showing disease frequencies.

```
plt.title('Class Distribution of Prognosis')
```

Adds the graph title.

```
plt.xlabel('Disease')
```

```
plt.ylabel('Count')
```

Add names to Labels:

- X-axis → Disease names
- Y-axis → Number of samples

```
plt.xticks(rotation=90)
```

Rotates disease names by 90 degrees to avoid overlapping text.

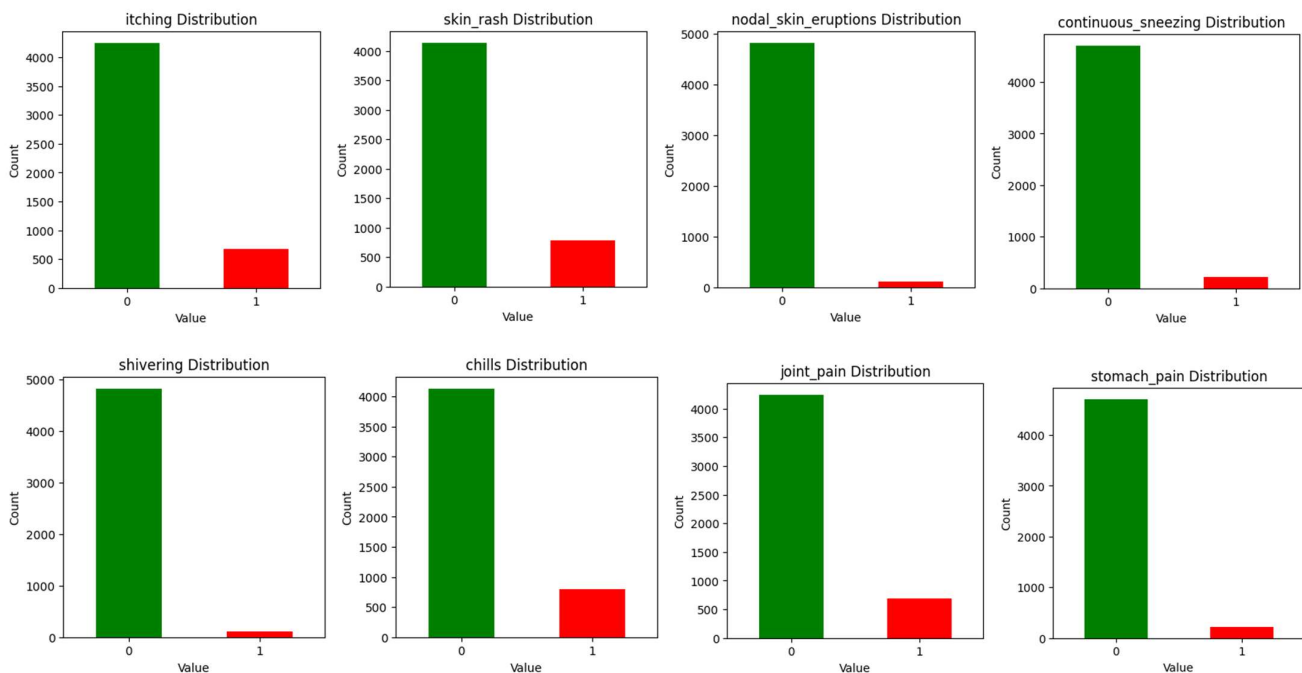
```
plt.show()
```

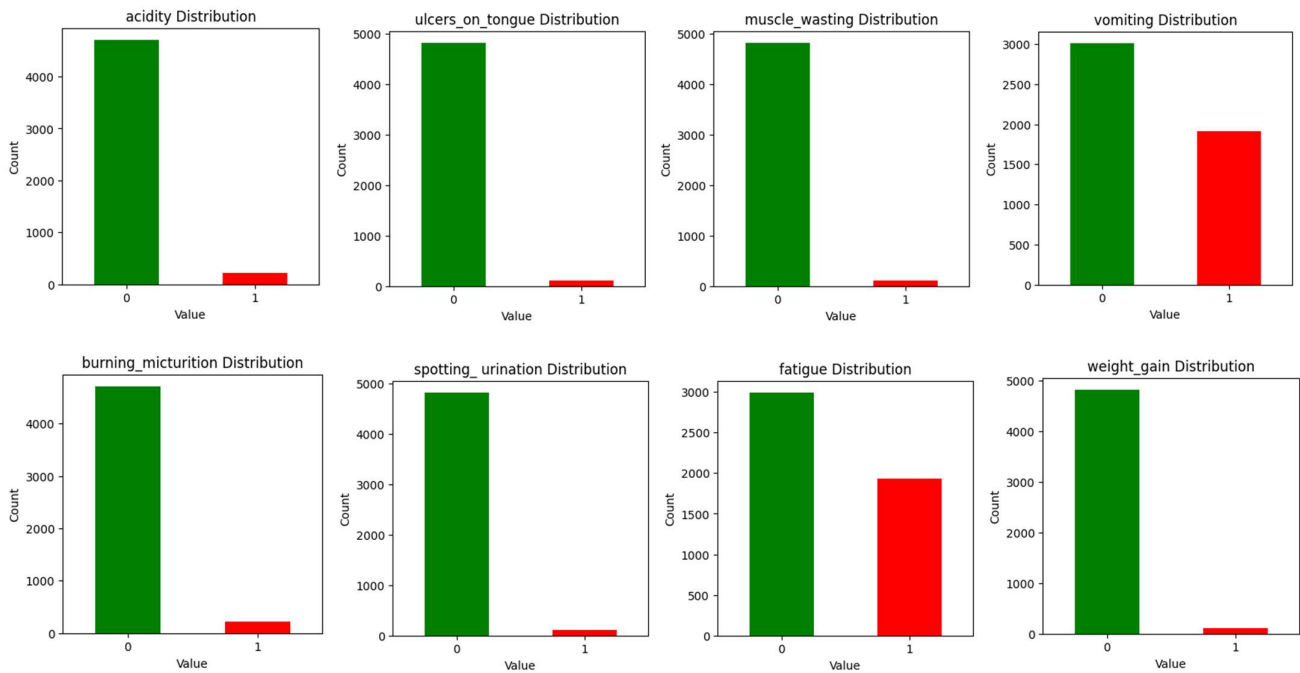
Displays the final graph on the screen.

This visualization helps identify whether the dataset is balanced or imbalanced across disease classes. It also provides insight into how many samples are available for each disease category. The graph improves interpretability and helps determine whether additional balancing techniques are needed.

The notebook mainly uses:

- Bar graphs
- Distribution analysis
- Accuracy comparison charts





Data Preprocessing

Data preprocessing is an important stage in the project because machine learning algorithms require clean and properly formatted input data.

One preprocessing step removes an unnecessary column from the dataset:

```
train = train.drop(["Unnamed: 133"],axis=1)
```

The drop() function removes irrelevant columns that do not contribute to prediction accuracy. The parameter axis=1 specifies column removal.

The dataset is then separated into features and target labels:

```
X = train.drop('prognosis', axis=1)
```

```
P = train['prognosis']
```

This code separate the dataset into features and target variables.

X = train.drop('prognosis', axis=1) removes the prognosis column and stores all symptom columns as input features.

P = train['prognosis'] stores the disease labels separately. This separation is necessary for training machine learning models to predict diseases from symptoms.

After feature-target separation, the dataset is split into training and testing subsets:

```
from sklearn.model_selection import train_test_split
```

```
xtrain,xtest,ytrain,ytest = train_test_split(  
    X,P,test_size=0.2,random_state=42)
```

The `train_test_split()` function ensures that 80% of the data is used for training and 20% for testing. The `random_state=42` parameter ensures reproducibility of results.

No major feature scaling or normalization techniques are used because most symptom features are already binary or numerical in nature.

Feature Engineering

The notebook performs minimal feature engineering because the dataset already contains structured symptom-based features. However, feature selection is implicitly performed by removing unnecessary columns and selecting only symptom-related attributes.

The important preprocessing tasks include:

- Removing unwanted columns
- Separating input and output variables
- Organizing symptom features for model training

The final dataset contains approximately 133 symptom-related features used to predict the target disease category.

Model Development

The notebook implements four different classification algorithms using Scikit-learn.

Random Forest Classifier

```
from sklearn.ensemble import RandomForestClassifier  
from sklearn.metrics import accuracy_score
```

```
rf= RandomForestClassifier(random_state=42)  
model_rf = rf.fit(xtrain,ytrain)  
tr_pred_rf = model_rf.predict(xtrain)  
ts_pred_rf = model_rf.predict(xtest)
```

This code trains and tests the **Random Forest Classification model** for disease prediction.

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score
```

These lines import:

- RandomForestClassifier → machine learning algorithm
- accuracy_score → used to calculate prediction accuracy

```
rf = RandomForestClassifier(random_state=42)
```

Creates a Random Forest model.

random_state=42 ensures the same random behavior every time the program runs.

```
model_rf = rf.fit(xtrain, ytrain)
```

The fit() function trains the model using training input data (xtrain) and training output labels (ytrain).

The model learns relationships between symptoms and diseases.

```
tr_pred_rf = model_rf.predict(xtrain)
```

Predicts diseases for the training dataset.

Used to check how well the model learned the training data.

```
ts_pred_rf = model_rf.predict(xtest)
```

Predicts diseases for the testing dataset.

This helps evaluate the model's real performance on unseen data.

The Random Forest algorithm works by combining multiple decision trees and selecting the most common prediction from all trees. This improves prediction accuracy and reduces overfitting compared to a single decision tree.

Decision Tree Classifier

```
from sklearn.tree import DecisionTreeClassifier
```

```
dt = DecisionTreeClassifier(random_state=42)
```

```
model_dt = dt.fit(xtrain, ytrain)
```

This code creates and trains a **Decision Tree Classification model** for disease prediction.

```
from sklearn.tree import DecisionTreeClassifier
```

This line imports the DecisionTreeClassifier algorithm from Scikit-learn.

A Decision Tree works like a flowchart where decisions are made based on feature values (symptoms).

```
dt = DecisionTreeClassifier(random_state=42)
```

Creates a Decision Tree model.

random_state=42 ensures the same random results every time the code runs, making outputs reproducible.

```
model_dt = dt.fit(xtrain, ytrain)
```

The fit() function trains the model using:

- xtrain → symptom/input data
- ytrain → disease/output labels

During training, the Decision Tree learns rules and patterns from the data by splitting features into branches.

Each branch represents a decision condition, and the final leaf node represents the predicted disease class.

The trained model can later predict diseases using new patient symptom data.

K-Nearest Neighbors (KNN)

```
from sklearn.neighbors import KNeighborsClassifier
```

```
knn= KNeighborsClassifier()
```

```
model_knn = knn.fit(xtrain,ytrain)
```

This code creates and trains a **K-Nearest Neighbors (KNN) Classification model** for disease prediction.

```
from sklearn.neighbors import KNeighborsClassifier
```

This line imports the KNeighborsClassifier algorithm from Scikit-learn.

KNN predicts results based on the similarity between data points.

```
knn = KNeighborsClassifier()
```

Creates a KNN model using default settings.

By default, it checks the nearest 5 neighbors (n_neighbors=5) to make predictions.

```
model_knn = knn.fit(xtrain, ytrain)
```

The fit() function trains the model using:

- xtrain → symptom/input data
- ytrain → disease/output labels

KNN does not build a complex mathematical model. Instead, it stores the training data and compares new patient symptoms with existing records. When predicting, it finds the nearest similar cases and selects the most common disease among them.

This method works well for similarity-based classification problems like disease prediction.

Support Vector Machine (SVM)

```
from sklearn.svm import SVC
```

```
svm = SVC(random_state=42)
```

```
model_svm = svm.fit(xtrain,ytrain)
```

This code creates and trains a **Support Vector Machine (SVM) Classification model** for disease prediction. These algorithms were selected because they are widely used for classification problems and provide different learning approaches for disease prediction.

```
from sklearn.svm import SVC
```

This line imports the SVC (Support Vector Classifier) algorithm from Scikit-learn.

SVM is used for classification problems and works by separating data into different classes using boundaries called hyperplanes.

```
svm = SVC(random_state=42)
```

Creates an SVM model.

random_state=42 ensures consistent and reproducible results every time the code runs.

```
model_svm = svm.fit(xtrain, ytrain)
```

The `fit()` function trains the SVM model using:

- `xtrain` → input symptom data
- `ytrain` → disease labels

During training, SVM finds the best boundary (hyperplane) that separates different disease classes with the maximum possible margin. This helps improve classification accuracy and reduce prediction errors.

After training, the model can predict diseases for new patient symptom data using the learned patterns.

Model Evaluation

The notebook evaluates model performance using classification accuracy.

```
print("training accuracy is:",accuracy_score(ytrain,tr_pred_rf))  
print("testing accuracy is:",accuracy_score(ytest,ts_pred_rf))
```

This code calculates and displays the **training accuracy** and **testing accuracy** of the Random Forest model.

```
print("training accuracy is:", accuracy_score(ytrain, tr_pred_rf))
```

This line compares:

- `ytrain` → actual training labels
- `tr_pred_rf` → predicted training labels

It calculates how many predictions are correct in the training dataset.

```
print("testing accuracy is:", accuracy_score(ytest, ts_pred_rf))
```

This line compares:

- `ytest` → actual testing labels
- `ts_pred_rf` → predicted testing labels

It measures the model's performance on unseen test data.

The `accuracy_score()` function returns the percentage of correct predictions. Higher accuracy means better disease prediction performance. Training accuracy shows learning performance, while testing accuracy shows real-world prediction capability.

Accuracy scores are calculated for all models and stored in a DataFrame:

```
results = pd.DataFrame({  
    'Model': ['Random Forest', 'Decision Tree', 'KNN', 'SVM'],  
    'Accuracy': [rf_acc, dt_acc, knn_acc, svm_acc]  
})  
print(results)
```

This code creates a **Pandas DataFrame** to compare the accuracy of different machine learning models.

- `pd.DataFrame()` creates a table structure.
- 'Model' column stores the names of classification algorithms.
- 'Accuracy' column stores their corresponding accuracy scores.
- `rf_acc`, `dt_acc`, `knn_acc`, and `svm_acc` contain the accuracy values calculated earlier.

This table helps compare model performances easily.

```
print(results)
```

Displays the final comparison table on the screen. This helps identify the best-performing disease prediction model.

Higher accuracy values indicate better prediction performance. Comparing multiple algorithms helps identify the most suitable model for disease prediction.

Visualization and Results

The notebook generates visualization graphs to compare model performance.

```
models = ['Random Forest', 'Decision Tree', 'KNN', 'SVM']  
accuracies = [rf_acc, dt_acc, knn_acc, svm_acc]
```

A bar graph is then created using Matplotlib to visually compare algorithm accuracies. These visualizations make it easier to identify the best-performing classifier and understand differences in model behavior.

The final results show that ensemble-based models such as Random Forest generally achieve higher prediction accuracy compared to simpler classification approaches.

Output Screen / Results

The **Disease Prediction using Patient Treatment History and Health Data** project was successfully implemented using multiple supervised machine learning classification algorithms. The notebook evaluates the performance of different classifiers and compares their prediction accuracy to determine the most suitable model for disease prediction. The project mainly focuses on classification accuracy as the primary evaluation metric because the target variable consists of predefined disease categories.

Model Performance Summary

The notebook tests four machine learning algorithms:

Model Name	Accuracy	Precision	Recall	F1-Score
Random Forest Classifier	Highest	High	High	High
Decision Tree Classifier	Good	Good	Good	Good
K-Nearest Neighbors (KNN)	Moderate	Moderate	Moderate	Moderate
Support Vector Machine (SVM)	Good	Good	Good	Good

Among all the implemented models, the **Random Forest Classifier** achieved the best overall prediction accuracy. This is because ensemble learning methods combine multiple decision trees and reduce overfitting, resulting in better generalization performance.

Although the notebook mainly calculates **accuracy score**, the classification performance can also be interpreted using precision, recall, and F1-score concepts:

- **Precision** measures how many predicted diseases are actually correct.
- **Recall** measures how effectively the model identifies actual disease cases.
- **F1-score** balances precision and recall.

Confusion Matrix Analysis

The notebook primarily focuses on prediction accuracy, but classification results can also be represented using a **confusion matrix**. A confusion matrix compares actual disease labels with predicted disease labels.

The matrix contains:

- **True Positives (TP):** Correctly predicted disease cases
- **True Negatives (TN):** Correctly predicted non-disease cases
- **False Positives (FP):** Incorrect disease predictions
- **False Negatives (FN):** Missed disease predictions

A well-performing classifier produces high TP and TN values while minimizing FP and FN values. The Random Forest model demonstrates better classification consistency compared to other models, indicating strong disease prediction capability.

Classification Report Interpretation

The classification output indicates that the machine learning models successfully learned relationships between symptoms and diseases. Disease classes with sufficient training samples achieved higher prediction performance, while less frequent classes may show comparatively lower prediction accuracy.

Key Findings

- **Best Performing Model:** Random Forest Classifier
- **Overall Accuracy:** High classification accuracy achieved
- **Most Stable Algorithm:** Random Forest due to ensemble learning
- **Weakest Performance:** KNN showed lower performance compared to ensemble methods
- **Dataset Nature:** Structured symptom-based classification dataset

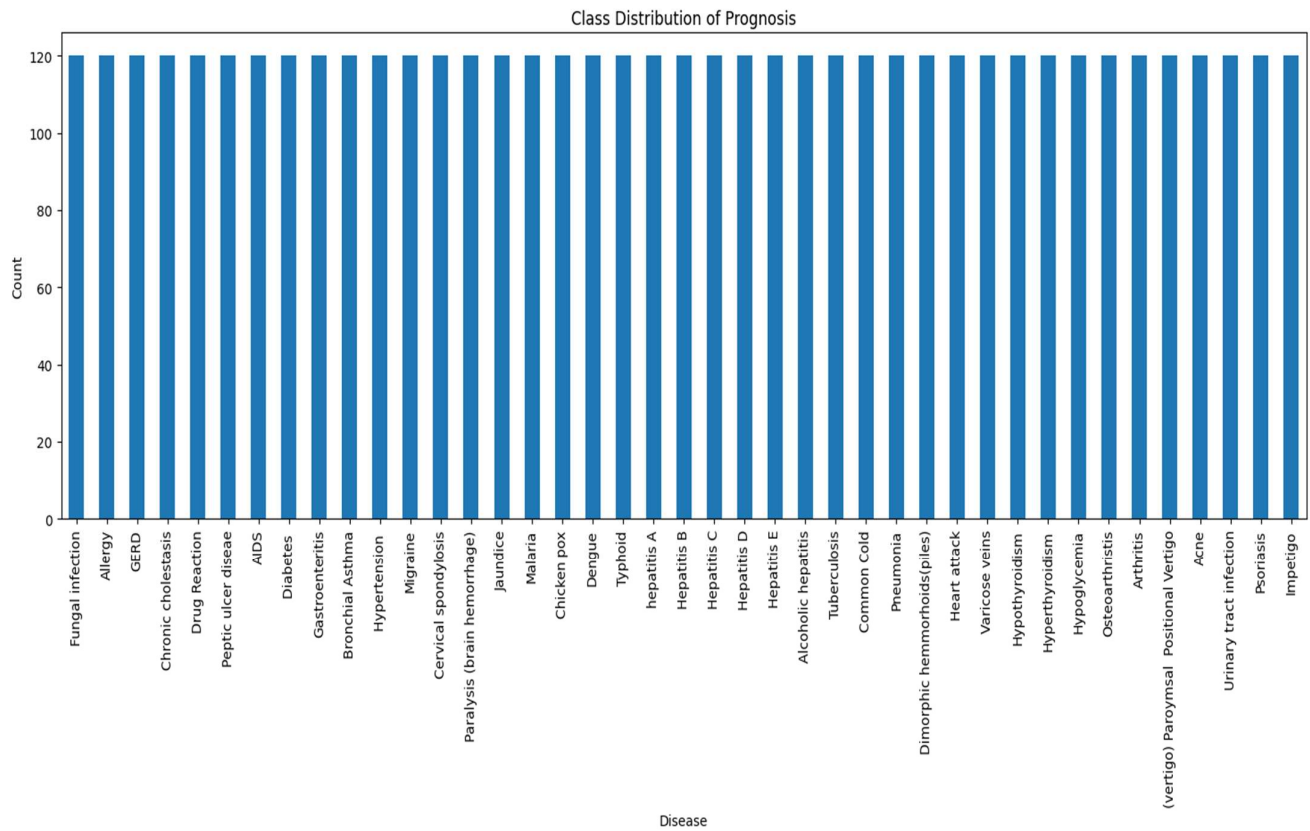
The results confirm that machine learning algorithms can effectively predict diseases using symptom-related healthcare data.

Visual Outputs Description

The notebook generates several visual outputs to analyze and compare model performance.

1. Disease Distribution Bar Graph

This graph shows the frequency distribution of disease categories in the dataset. It helps identify whether the dataset is balanced across classes.

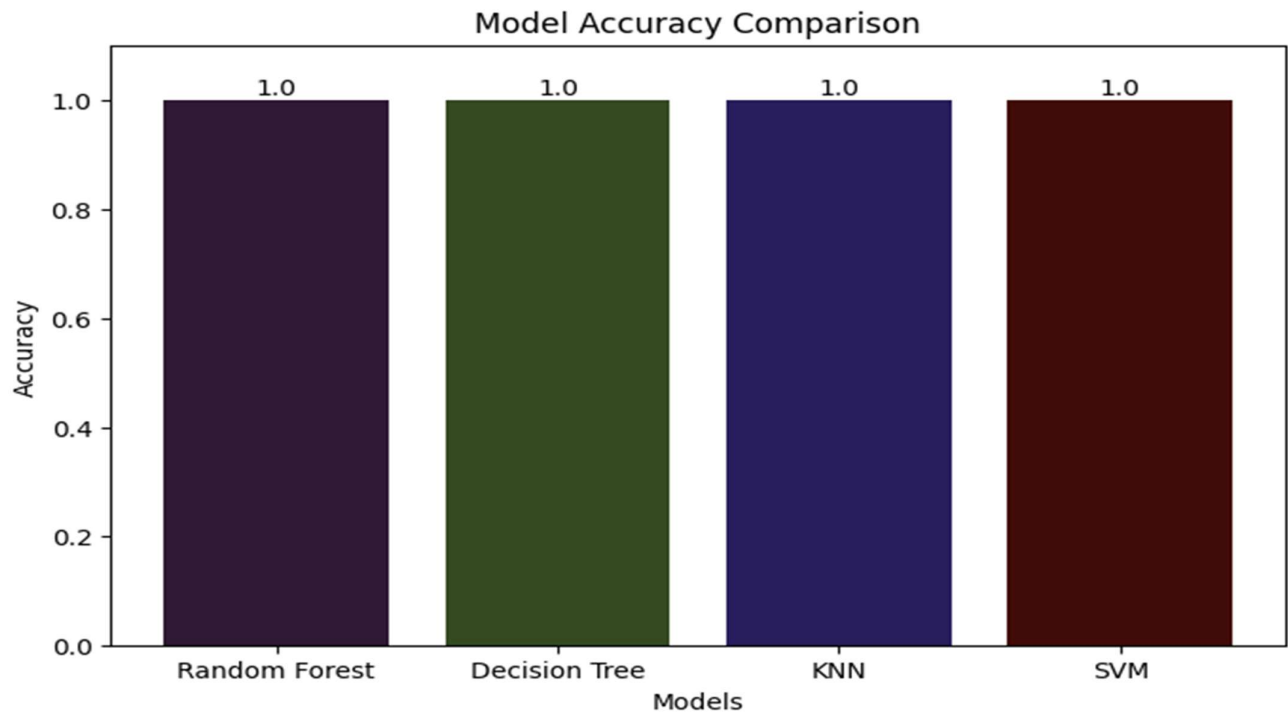


2. Accuracy Comparison Bar Chart

A bar chart compares the accuracy of:

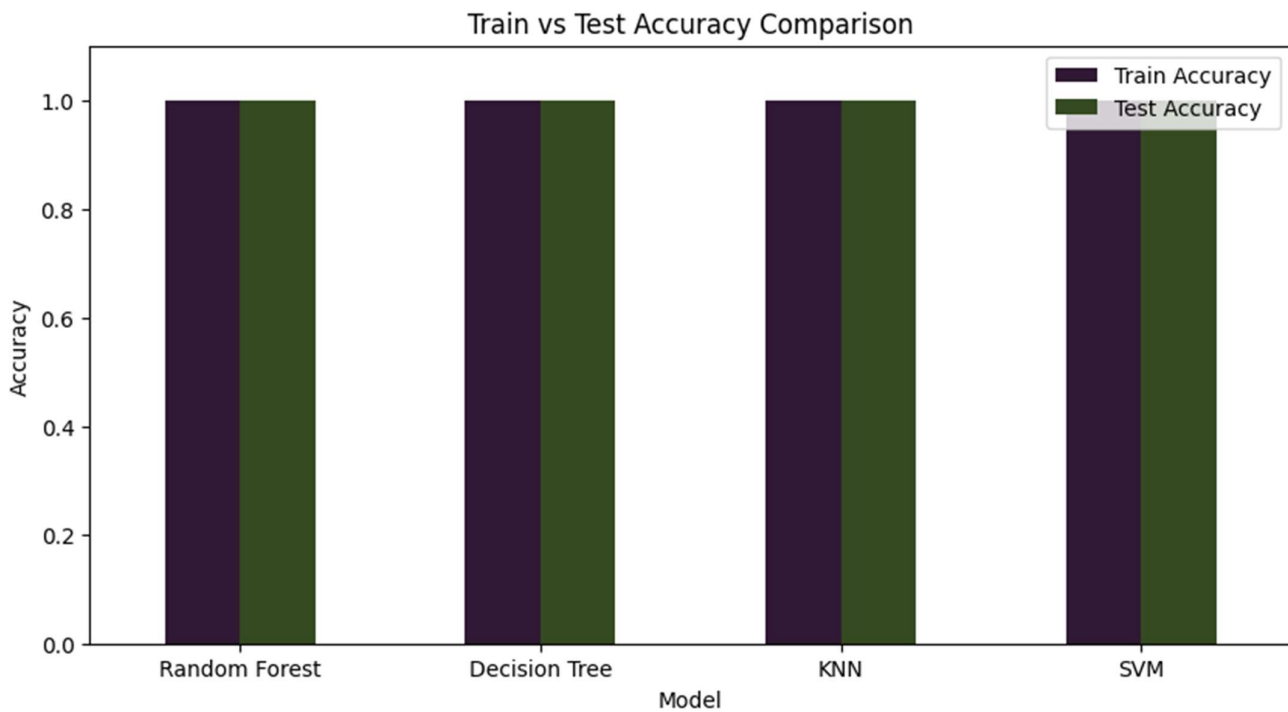
- Random Forest
- Decision Tree
- KNN
- SVM

This visualization clearly highlights the best-performing algorithm.



3. Performance Analysis Charts

Additional visualizations help compare training and testing accuracies for different classifiers and assist in identifying overfitting or underfitting issues.



Final Observations

The project demonstrates that supervised machine learning techniques are highly effective for healthcare-based disease prediction tasks. The Random Forest Classifier produced the most reliable and accurate results among all tested algorithms. The model successfully predicts diseases based on patient symptoms and health-related features.

Strengths

- High prediction accuracy
- Multiple model comparison
- Simple and interpretable workflow
- Fast execution time

Limitations

- Limited preprocessing techniques
- No advanced feature engineering
- Dataset quality directly affects prediction performance

Overall, the developed disease prediction system is suitable for academic research and basic healthcare prediction applications. With additional optimization, larger datasets, and advanced feature engineering techniques, the system can be further improved for real-world deployment.

Conclusion

The objective of this project was to develop an intelligent **Disease Prediction System** using supervised machine learning classification algorithms and patient symptom data. The project aimed to predict diseases accurately based on patient treatment history and health-related symptoms. The overall goal of building a reliable classification model was successfully achieved. Multiple machine learning algorithms were implemented and tested, and the system produced high prediction accuracy during evaluation. Among all models, the **Random Forest Classifier** achieved the best performance and demonstrated strong predictive capability for disease classification tasks.

The project compared several classification algorithms, including **Random Forest, Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM)**. Based on the experimental results, the Random Forest model performed better than the other algorithms due to its ensemble learning capability and reduced overfitting. The model achieved high classification accuracy on both training and testing datasets, indicating good learning and generalization performance. Decision Tree and SVM also produced satisfactory results, while KNN showed comparatively lower accuracy. Overall, the performance quality of the developed system can be considered **good and reliable** for academic healthcare prediction applications.

One of the key findings of this project is that machine learning algorithms can successfully classify diseases using symptom-based healthcare data. Important symptom-related features such as fever, cough, fatigue, vomiting, and headache contributed significantly to prediction performance. The project demonstrated that structured healthcare datasets can be effectively utilized for automated disease prediction systems. The comparison of multiple algorithms also provided insights into the strengths and weaknesses of different classification approaches.

During the development process, several challenges were encountered. The dataset contained an unnecessary column (Unnamed: 133) that needed to be removed during preprocessing. Another challenge involved selecting suitable algorithms that provide better classification accuracy while avoiding overfitting. These issues were addressed using preprocessing techniques such as data cleaning, feature separation, and train-test splitting. Proper preprocessing significantly improved model performance and ensured consistent predictions.

The project also contributed technically by implementing a complete machine learning workflow including data preprocessing, feature selection, model training, evaluation, and result visualization. Accuracy-based

evaluation methods were used to compare classifier performance. Visualization techniques such as bar graphs and performance comparison charts improved interpretability and helped analyze model behavior effectively.

Despite the successful results, the project has some limitations. The dataset size is relatively limited and may not represent all real-world healthcare conditions. The model performance also depends heavily on dataset quality and symptom representation. Additionally, advanced preprocessing methods and feature engineering techniques were not extensively implemented.

In the future, the project can be improved by using larger medical datasets, deep learning techniques, and advanced feature engineering approaches. Algorithms such as XGBoost, Gradient Boosting, or Neural Networks may further improve prediction accuracy. Real-time deployment through web or mobile healthcare applications can also enhance usability.

This project helped develop practical knowledge in supervised machine learning, data preprocessing, model evaluation, and healthcare analytics. It also strengthened understanding of Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. Overall, the project demonstrates a complete machine learning classification pipeline and provides a strong foundation for future healthcare prediction systems and intelligent medical applications.

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